Functional Annotation of Mouse *Scgb3a2* (Secretoglobin family 3A member 2 / UGRP1 / PnSP-1; UniProt Q920H1)

Summary

Mouse **Scgb3a2** encodes **Secretoglobin family 3A member 2 (SCGB3A2)**, also known historically as **Uteroglobin-Related Protein 1 (UGRP1)** and **Pneumo Secretory Protein 1 (PnSP-1)**. It is a small (~10 kDa) **secreted protein** that assembles into a disulfide-linked **homodimer** (and higher-order tetramer) and belongs to the **secretoglobin superfamily, UGRP subfamily** — the same structural family as uteroglobin/Clara cell secretory protein. Gene identity, organism (*Mus musculus*), and protein family all match the UniProt record precisely; the literature reviewed is unambiguously about this gene, so no identity conflict exists. SCGB3A2 is **not an enzyme, transporter, or structural/cytoskeletal protein**. Instead, it functions extracellularly as a **cytokine-/growth-factor-like immunomodulator and a secreted lipid/ligand-binding chaperone**.

The protein is produced almost exclusively by **airway epithelial Club (Clara) cells** of the lung and trachea, under the direct transcriptional control of the homeodomain transcription factor **NKX2-1 (T/EBP/TITF1)**. From its site of secretion into the airway lumen and alveolar space, SCGB3A2 carries out four interlocking, well-documented activities: (1) an **anti-inflammatory** activity that suppresses Th2-driven allergic airway inflammation by **inhibiting the NF-κB pathway**; (2) a **growth-factor** activity that promotes **fetal lung development and maturation**; (3) a **tissue-protective/anti-fibrotic and anti-emphysema** activity mediated by **STAT1/STAT3 signaling that transcriptionally upregulates α1-antitrypsin (Serpina1a)**; and (4) an **innate-immune and anti-cancer** activity in which SCGB3A2 acts as an **LPS chaperone**, binding bacterial lipopolysaccharide and delivering it into the cytosol via the cell-surface receptor **syndecan-1** to trigger **caspase-11–driven pyroptosis**. It also binds the alveolar-macrophage scavenger receptor **MARCO** and recognizes microbial surfaces, linking it to pathogen clearance.

The coherent picture that emerges is of a Club-cell secretoglobin that operates at the air–epithelium interface as a multifunctional signaling and chaperone molecule, using **two identified receptors (MARCO and syndecan-1)** and converging on **two principal signaling outputs (NF-κB inhibition and STAT3 activation)** to defend and remodel the lung. This report lays out the evidence for each of these functions, synthesizes them into a mechanistic model, evaluates the literature, and identifies the remaining knowledge gaps.

Key Findings

Finding 1 — Identity: SCGB3A2/UGRP1 is a secreted, NKX2-1–controlled homodimeric secretoglobin of airway Club cells

SCGB3A2 (gene *Scgb3a2*; synonyms *Ugrp1*, *Pnsp1*) was originally discovered as a novel gene **down-regulated in the lungs of T/ebp/Nkx2.1-null mouse embryos**, identifying it from the outset as a downstream target of the master lung/thyroid transcription factor NKX2-1. The gene has a compact structure of **3 exons and 2 introns** and maps to mouse **chromosome 18C-D**, a region homologous to **human 5q31-34** — a locus repeatedly linked to asthma and atopic phenotypes. The gene product is a **~10 kDa secreted protein** that forms a **homodimer** (and tetramer) and shows amino-acid sequence similarity to the **uteroglobin/Clara cell secretory protein family** of secretoglobins (PMID: 11682631).

Expression is **predominantly in the lung** (airway epithelial Club/Clara-like cells) and trachea, with only low levels in thyroid. Developmentally, expression is detectable from early organogenesis and rises markedly by E16.5. Crucially, **cotransfection/reporter assays confirmed that T/EBP/NKX2.1 directly regulates the *Ugrp1* promoter** at the transcriptional level (PMID: 11682631). This establishes the cell of origin, the secreted and dimeric nature of the protein, its family membership, and its upstream transcriptional regulator — the foundational facts on which all functional interpretations rest.

> *"The gene product is a secreted protein, forms a homodimer, and exhibits an amino acid sequence similar to that seen in the uteroglobin/Clara cell secretory protein family of proteins"* — PMID: 11682631

Finding 2 — Anti-inflammatory activity: suppression of allergic airway inflammation via NF-κB inhibition

SCGB3A2 is a bona fide **anti-inflammatory factor** in the airway. When adenovirus-delivered UGRP1 was expressed in the airway epithelium of ovalbumin (OVA)-sensitized and -challenged mice, it **markedly reduced the number of infiltrating inflammatory cells, particularly eosinophils**, in lung tissue and lowered the Th2 cytokines **IL-4, IL-5, and IL-13** in bronchoalveolar lavage (BAL) fluid (PMID: 16456148). Mechanistically, more recent work shows that SCGB3A2 overexpression **mitigates toluene-diisocyanate (TDI)-induced inflammation and airway remodeling by inhibiting the NF-κB pathway and enhancing autophagy**; critically, re-activating NF-κB abrogated the protective effect, demonstrating that NF-κB inhibition is causal, not incidental (PMID: 40865253).

SCGB3A2 sits within an **anti-inflammatory regulatory loop**: its own expression is suppressed during active inflammation and induced by resolving signals. *Ugrp1* mRNA is strongly inversely correlated with IL-5 ($r = 0.948$, $P < 0.0001$) and with IL-9 ($r = 0.616$, $P = 0.023$) in BAL, and intranasal IL-5 or IL-9 reduces lung *Ugrp1* (PMID: 15626484; PMID: 15531759). Conversely, the anti-inflammatory cytokine **IL-10 induces *UGRP1* transcription through two NKX2-1 (T/EBP) binding sites** in the promoter (PMID: 15485815). Thus SCGB3A2 is both an effector and a readout of the anti-inflammatory state of the airway.

> *"markedly reduced the number of infiltrating inflammatory cells, particularly eosinophils, in lung tissue as well as the level of proinflammatory cytokines such as interleukin (IL)-4, IL-5, and IL-13 in bronchoalveolar lavage fluids"* — PMID: 16456148

Finding 3 — Growth-factor activity: promotion of fetal lung development and maturation

SCGB3A2 behaves as a **growth factor for the developing lung**. Its expression rises sharply around **E16.5**, precisely when the lung undergoes dramatic morphogenetic changes. In *ex vivo* organ culture, SCGB3A2 **promoted early lung development of both wild-type and Nkx2-1-null fetal lungs**. Administered intravenously to Nkx2-1-heterozygous pregnant dams, it **rescued/advanced organogenesis in null fetuses**, and given to pregnant wild-type mice during mid-to-late organogenesis it **significantly promoted late lung maturation**, as judged by fetal breathing motion, gross/histological lung morphology, gestational-stage-specific gene expression, and surfactant phospholipid profiles (PMID: 18535256).

> *"SCGB3A2 significantly promoted both early and late stages of lung dev"* — PMID: 18535256

This activity is distinct from simple anti-inflammation: it is a direct, dose-responsive developmental/maturational effect, consistent with the protein's sharply timed developmental expression.

Finding 4 — Anti-fibrotic activity and MARCO binding (innate defense)

Endogenous SCGB3A2 **protects against pulmonary fibrosis**. In a loss-of-function model, **Scgb3a2-null mice given bleomycin developed more severe fibrosis than wild-type**, with greater weight loss, higher Ashcroft fibrosis scores, elevated hydroxyproline and collagen mRNAs, more BAL inflammatory cells, and higher IL-4 and TGFβ mRNAs (PMID: 26559674). This genetic evidence establishes that the protein normally restrains TGFβ-driven fibrotic remodeling.

Separately, a key mechanistic study identified a receptor and a microbial-recognition role. Secreted recombinant UGRP1 co-purified with apolipoprotein A-I, and expression cloning identified the **macrophage scavenger receptor with collagenous structure (MARCO)** — expressed by alveolar macrophages — as a **UGRP1 receptor**. UGRP1 also **bound bacteria and yeast**, and **LPS competed with UGRP1 for binding to MARCO and to bacteria** (PMID: 12847263). This defines a lung-specific ligand–receptor pair and places SCGB3A2 squarely within innate immune surveillance and pathogen clearance.

> *"BLM-treated Scgb3a2-null mouse lungs had more severe fibrosis than those of wild-type mouse lungs"* — PMID: 26559674
>
> *"we identified macrophage scavenger receptor with collagenous structure (MARCO) as a receptor for UGRP1"* — PMID: 12847263

Finding 5 — LPS chaperone: delivery of LPS via syndecan-1 to trigger caspase-11 pyroptosis

The most mechanistically precise function assigned to SCGB3A2 is that of an **LPS-delivery chaperone**. Secreted SCGB3A2 **binds lipopolysaccharide (LPS) and chaperones it into the cytosol through the cell-surface receptor syndecan-1**, thereby activating the **non-canonical inflammasome and caspase-11–driven pyroptosis** (PMID: 30526845). SCGB3A2 + LPS co-treatment significantly induced pyroptosis of RAW264.7 macrophages and decreased cancer-

cell proliferation *in vitro*, and SCGB3A2 treatment **reduced progression of xenograft tumors in mice**. The authors conclude that SCGB3A2 is a **critical LPS-delivery vehicle** that unifies its innate-immune and anti-cancer activities.

> *"secretoglobin(SCGB)3A2, a small protein predominantly secreted in airways, chaperones LPS to the cytosol through the cell surface receptor syndecan-1; this leads to pyroptotic cell death driven by caspase-11"* — PMID: 30526845

This finding is especially important because it provides a concrete **molecular mechanism** — ligand (LPS), receptor (syndecan-1), route (cytosolic delivery), and effector (caspase-11/pyroptosis) — for a secretoglobin whose family is best known for ligand-binding hydrophobic cavities.

Finding 6 — STAT3 → α1-antitrypsin axis protects the lung from emphysema

SCGB3A2 protects alveolar structure through a defined transcriptional cascade. In a 6-month cigarette-smoke COPD model, **Scgb3a2-knockout mice lost lung structure even under control conditions** and developed more airspace expansion and alveolar-wall destruction than wild-type after smoke exposure, whereas **Scgb3a2-overexpressing (transgenic) lungs were protected** (PMID: 36796505). Mechanistically, SCGB3A2 **increased the expression and phosphorylation of STAT1 and STAT3 and increased α1-antitrypsin (A1AT) expression** in mouse lung fibroblast (MLg) and epithelial (MLE-15) cells. A1AT levels fell with *Stat3* knockdown and rose with *Stat3* overexpression; SCGB3A2 stimulation induced **STAT3 homodimer formation**; and **ChIP and reporter assays showed STAT3 binds specific sites on *Serpina1a* (encoding A1AT) and upregulates its transcription**, with nuclear localization of phosphorylated STAT3.

> *"SCGB3A2 increased the expression and phosphorylation of signal transducers and activators of transcription (STAT)1 and STAT3, and the expression of α1-antitrypsin (A1AT) in mouse lung fibroblast-derived MLg cells and mouse lung epithelial-derived MLE-15 cells"* — PMID: 36796505
>
> *"STAT3 binds to specific binding sites on the Serpina1a gene encoding A1AT and upregulates its transcription in lung tissues of mice"* — PMID: 36796505

Because α1-antitrypsin is the principal inhibitor of neutrophil elastase and its deficiency is a classic cause of emphysema, this axis provides a mechanistic explanation for the protein's anti-emphysema and tissue-protective activities.

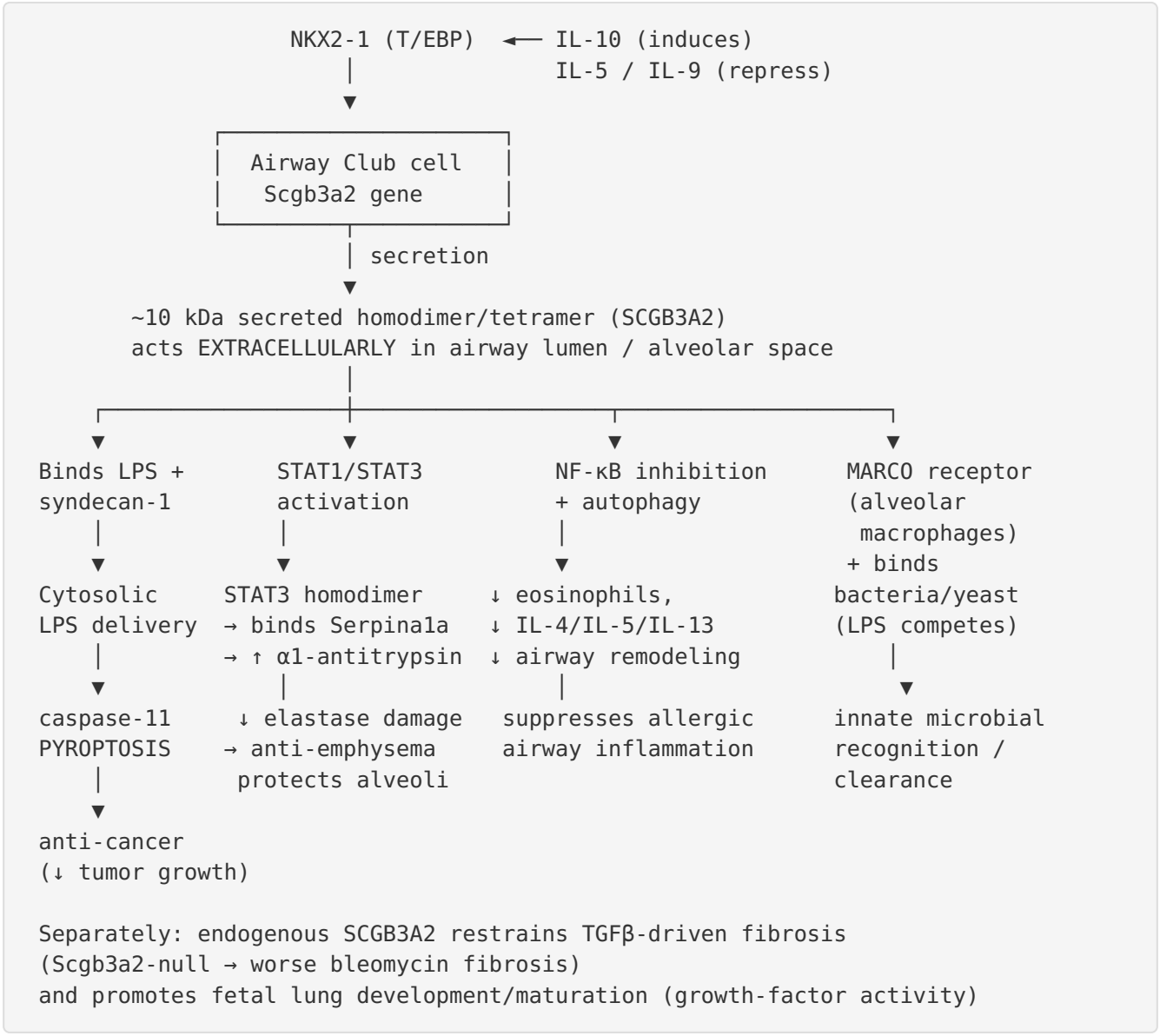
Finding 7 — Integrated model: a secreted secretoglobin with two receptors and two signaling outputs, not an enzyme

The authoritative review by Kimura et al. (2022) consolidates these threads: SCGB3A2 is *"a small secretory protein of ~10 kDa that forms a dimer and a tetramer... predominantly expressed in airway epithelial Club cells, and has anti-inflammatory, growth factor, anti-fibrotic, and anti-cancer activities that influence various lung diseases"* (PMID: 35016921). Across the primary literature these resolve into anti-inflammatory activity via NF-κB inhibition, growth-factor/tissue-protective activity via STAT1/STAT3 → Serpina1a, and innate-immune/anti-cancer activity via LPS chaperoning through syndecan-1 and MARCO binding. **No catalytic/enzymatic or transporter activity has been reported.**

> *"SCGB3A2 is a small secretory protein of ~10 kDa that forms a dimer and a tetramer. SCGB3A2 is predominantly expressed in airway epithelial Club cells, and has anti-inflammatory, growth factor, anti-fibrotic, and anti-cancer activities that influence various lung diseases"* — PMID: 35016921

Mechanistic Model / Interpretation

SCGB3A2 is best understood as a **Club-cell secretoglobin that functions at the air–epithelium interface as a multifunctional signaling and chaperone protein**. The following schematic integrates the findings:



Two receptors, two outputs. The protein engages at least two cell-surface receptors: **MARCO** on alveolar macrophages (microbial recognition) and **syndecan-1** (cytosolic LPS delivery). Its intracellular signaling converges on two principal transcriptional outputs: **inhibition of NF-κB** (anti-inflammatory arm) and **activation of STAT1/STAT3** (tissue-protective/growth-factor arm, with *Serpina1a*/A1AT as the defined target gene).

A unifying biochemical theme — ligand binding, not catalysis. The secretoglobin fold is characterized by a hydrophobic internal cavity (classically demonstrated for uteroglobin's progesterone binding; PMID: 7770456, and for the uteroglobin-like cat allergen Fel d 1; PMID: 17543334). SCGB3A2 exploits this ligand-binding capacity to **bind and chaperone LPS** (and to

bind bacterial/fungal surfaces), rather than to catalyze any reaction. This reframes several of its "activities" as downstream consequences of a core **ligand-binding/chaperone** biochemical function combined with receptor-mediated signaling.

Activity	Trigger / Receptor	Signaling output	Functional outcome	Key evidence
Anti-inflammatory	(airway epithelium)	NF-κB inhibition, ↑autophagy	↓ eosinophils, ↓ Th2 cytokines, ↓ remodeling	16456148, 40865253
Growth factor	(fetal lung)	— (promotes maturation)	↑ lung development, surfactant	18535256
Anti-emphysema / tissue-protective	fibroblasts, epithelium	STAT1/STAT3 → *Serpina1a*	↑ α1-antitrypsin, ↓ alveolar destruction	36796505
Anti-fibrotic	(endogenous)	↓ TGFβ, ↓ IL-4	↓ bleomycin fibrosis	26559674
Innate immune / anti-cancer	LPS + syndecan-1; MARCO	caspase-11	pyroptosis, ↓ tumor growth, microbial binding	30526845, 12847263

Localization. All evidence indicates that SCGB3A2 is **secreted and acts extracellularly** — in the airway lumen and alveolar space (where it contacts macrophages via MARCO), and at the cell surface of target cells (via syndecan-1), from which it is routed to the **cytosol** specifically for the LPS-pyroptosis function. It is synthesized as a precursor (signal peptide) consistent with the classical secretory pathway.

Evidence Base

PMID	Study	Contribution	Strength
11682631	UGRP1 as NKX2-1 target gene	Gene discovery, secreted homodimer, secretoglobin family, NKX2-1 regulation	Primary; foundational identity
16456148	UGRP1 suppresses allergic airway inflammation	Gain-of-function: ↓ eosinophils, ↓ Th2 cytokines	Primary; in vivo mouse
40865253	Curcumin/SCGB3A2 autophagy & NF-κB	Mechanism: NF-κB inhibition is causal	Primary; mechanistic
15485815	IL-10 induces UGRP1 via T/EBP sites	Places SCGB3A2 in anti-inflammatory loop	Primary; promoter dissection
15626484	IL-5 reduces UGRP1	Inverse regulation by Th2 cytokine (r=0.948)	Primary; correlation + intervention
15531759	IL-9 reduces UGRP1	Inverse regulation (r=0.616) + localization	Primary
18535256	SCGB3A2 in lung development	Growth-factor activity in vivo/ex vivo	Primary; developmental
26559674	SCGB3A2 anti-fibrotic	Loss-of-function: worse bleomycin fibrosis	Primary; knockout
12847263	UGRP1–MARCO ligand–receptor pair	Receptor identification, microbial binding	Primary; biochemical
30526845	LPS uptake via syndecan-1 → pyroptosis	LPS chaperone mechanism + anti-cancer	Primary; mechanistic
36796505	SCGB3A2 protects from emphysema	STAT3 → Serpina1a/A1AT axis; ChIP	Primary; knockout + transgenic + mechanism
35016921	Kimura et al. review (2022)	Authoritative synthesis of all activities	Review

Supporting structural context. Studies of related secretoglobins — uteroglobin's progesterone-binding hydrophobic cavity (PMID: 7770456) and the calcium-binding, ligand-cavity-bearing uteroglobin-like fold of Fel d 1 (PMID: 17543334) — support the inference that SCGB3A2's shared fold provides a ligand-binding cavity consistent with its LPS-chaperone role.

Human genetics / translational context. Human *UGRP1* lies at the asthma-linked 5q31 locus; a functional −112G/A promoter polymorphism that reduces transcription was associated with asthma in a Japanese population (PMID: 11813133) and with allergic rhinitis in Singapore Chinese (PMID: 21410962), though several other populations (Indian, Sicilian, German) found no association (PMID: 15591807, PMID: 18201431, PMID: 12915772). These genetic data are consistent with — but do not by themselves establish — the anti-inflammatory function; their mixed results reflect population heterogeneity rather than contradicting the mechanistic mouse data. SCGB3A2 is also a candidate biomarker in lung adenocarcinoma (PMID: 21937081, PMID: 29363881).

Limitations and Knowledge Gaps

1. **No atomic structure of SCGB3A2 itself.** The ligand-binding cavity and the physical basis of LPS binding are inferred from homology to uteroglobin and Fel d 1, not from an experimental SCGB3A2 structure. A crystal or cryo-EM structure (apo and LPS-bound) would confirm the chaperone mechanism.
2. **Receptor usage is incompletely mapped.** MARCO and syndecan-1 are identified, but whether a dedicated signaling receptor mediates the STAT1/STAT3 and NF-κB effects — and how receptor choice is partitioned between the anti-inflammatory and pyroptotic outcomes — is unresolved. The signal that triggers STAT3 phosphorylation downstream of SCGB3A2 has not been traced to a specific receptor.
3. **Apparently opposing immune outputs.** SCGB3A2 is anti-inflammatory (NF-κB inhibition, ↓ eosinophils) yet pro-pyroptotic (caspase-11 activation with LPS). The context (LPS availability, cell type, receptor engaged) that dictates which program dominates is not fully defined.
4. **Molecular-weight / oligomerization details.** Reports cite ~10 kDa (monomer) and ~17 kDa (homodimeric secretory protein in early human work); clarifying the physiological oligomeric state in vivo and whether tetramers are functionally distinct from dimers remains open.

5. **Human vs. mouse translation.** Most mechanistic data are from mouse; human genetic associations are inconsistent across populations. The degree to which the mouse mechanisms (especially the STAT3→*SERPINA1*/A1AT axis and syndecan-1 LPS delivery) operate in human airways needs direct confirmation.
6. **Endogenous ligand spectrum.** Beyond LPS and microbial surfaces, the full repertoire of hydrophobic/lipid ligands that the secretoglobin cavity binds (e.g., phospholipids, steroids) has not been systematically characterized for SCGB3A2.

Proposed Follow-up Experiments / Actions

1. **Determine the SCGB3A2 structure, apo and LPS-bound**, by X-ray crystallography or cryo-EM, and map LPS contacts by mutagenesis to test whether the secretoglobin cavity is the LPS-binding site.
2. **Define the signaling receptor(s) for the STAT3 arm.** Use proximity labeling (BioID/APEX) or affinity capture on lung fibroblasts/epithelial cells to identify the receptor upstream of SCGB3A2-induced STAT1/STAT3 phosphorylation, then validate with receptor knockdown/knockout and loss of A1AT induction.
3. **Dissect the anti-inflammatory vs. pyroptotic switch.** Titrate LPS and compare syndecan-1^{+} vs. syndecan-1^{-} and MARCO^{+} vs. MARCO^{-} cells to establish which receptor/context routes SCGB3A2 toward NF-κB inhibition versus caspase-11 pyroptosis.
4. **Test the STAT3→*Serpina1a* axis genetically in vivo.** Cross *Scgb3a2*-transgenic mice with *Stat3* or *Serpina1a* conditional alleles to confirm that A1AT induction is required for the anti-emphysema phenotype under cigarette-smoke exposure.
5. **Validate human translatability.** Measure SCGB3A2-dependent A1AT induction and syndecan-1–mediated LPS uptake in primary human airway epithelial cells and macrophages; correlate with human airway-disease outcomes.
6. **Characterize the ligand repertoire** of recombinant SCGB3A2 by lipidomic co-purification and biophysical binding assays to define substrate specificity of the chaperone cavity beyond LPS.
7. **Evaluate therapeutic potential** of recombinant SCGB3A2 (or stabilized dimer/tetramer) in preclinical models of asthma, COPD/emphysema, pulmonary fibrosis, and lung cancer, leveraging its convergent anti-inflammatory, tissue-protective, and anti-tumor activities.

Conclusion

Mouse SCGB3A2/UGRP1 (Q920H1) is a small, secreted, NKX2-1–driven **secretoglobin of airway Club cells** whose primary molecular function is **ligand binding/chaperoning and receptor-mediated immunomodulation — not enzymatic catalysis or transport**. Acting extracellularly in the airway and alveolar space, it suppresses allergic inflammation by inhibiting NF-κB, promotes fetal lung development, protects alveolar structure from emphysema through a STAT1/STAT3 → α1-antitrypsin (*Serpina1a*) transcriptional axis, restrains fibrosis, and serves innate host-defense and anti-cancer functions by binding microbial surfaces (via MARCO) and chaperoning LPS into the cytosol (via syndecan-1) to trigger caspase-11 pyroptosis. These activities are well supported by convergent gain-of-function, loss-of-function (knockout), and mechanistic (ChIP, reporter, receptor-identification) evidence in mouse models, synthesized in a 2022 authoritative review.